

Elliptic Curve over Reals

Definition

let $a, b \in \mathbb{R}$ such that

$$4a^3 + 27b^2 \neq 0$$

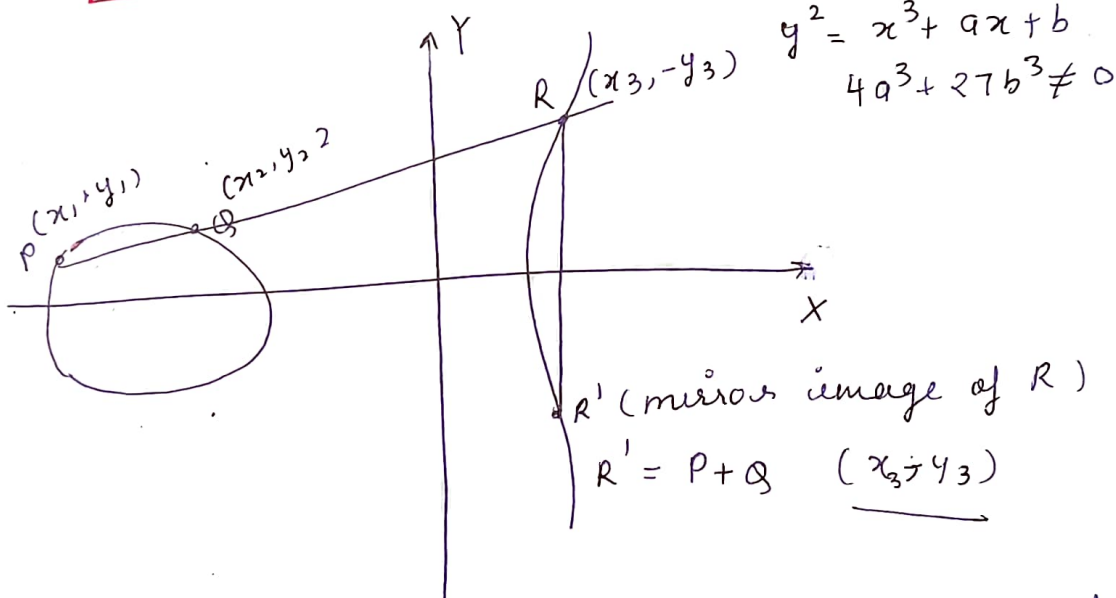
$$E = \left\{ (x, y) \mid y^2 = x^3 + ax + b \text{ and } a, b \in \mathbb{R} \cup \{\infty\} \right. \\ \left. \text{such that } 4a^3 + 27b^2 \neq 0 \right\}$$

point at infinity

3 real roots

Non-singular Elliptic curve

If $4a^3 + 27b^2 = 0$ - singular elliptic curve



Let's take two points P & Q on the curve draw a line PQ which touches another part of curve at R

Case 1 If P & Q are different

$$x_1 \neq x_2$$

λ = slope of PQ

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1}$$

eqn of PQ is $(y - y_1) = \lambda(x - x_1)$

$$(y - y_1) = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) \quad \text{line passing through } R$$

$$-y_3 - y_1 = \frac{(y_2 - y_1)}{x_2 - x_1} (x_3 - x_1) \quad (x_3, -y_3)$$

$$y_3 = \frac{y_1 - y_2}{x_1 - x_3} (x_3 - x_1) - y_1$$

$$\boxed{y_3 = \lambda (x_1 - x_3) - y_1} \quad \text{--- (1)}$$

$$y^2 = x^3 + ax + b \quad \text{--- } y = y_1 + \lambda (x - x_1)$$

put value of y_3

$$\boxed{(y_1 + \lambda (x - x_1))^2 = x^3 + ax + b}$$

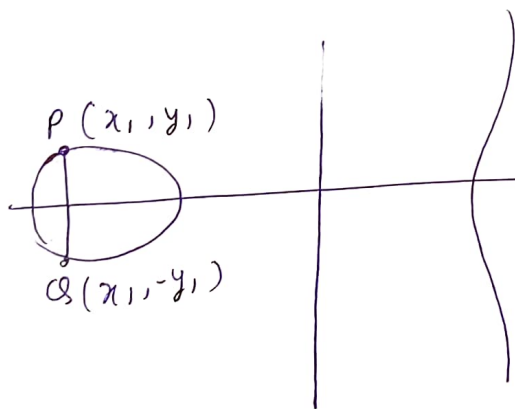
cubic in x and have 3 roots

$$x_1 + x_2 + x_3 = \lambda^2$$

$$\boxed{x_3 = \lambda^2 - (x_1 + x_2)} \quad \text{--- (2)}$$

$(E, '+')$ is a gp- θ identity element

Case 2 $x_1 = x_2, y_1 = -y_2$ (P & Q are mirror images of each other)



$$Q = -P$$

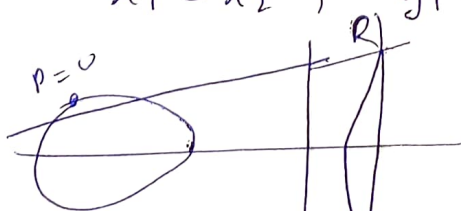
intersection will be at infinity

$$P + Q = \theta \text{ (point at infinity)}$$

$$P + (-P) = \theta$$

Case 3

$x_1 = x_2, y_1 = y_2$ (P, Q are same)



$$P + Q = P + P = (x_3, y_3)$$

$$x_3 = \lambda^2 - x_1 - x_2$$

$$= \lambda^2 - 2x_1$$

$$y_3 = \lambda(x_1 - x_3) - y_1$$

$$y^2 = x^3 + ax + b$$

$$2y \frac{dy}{dx} = 3x^2 + a$$

$$\frac{dy}{dx} = \frac{3x^2 + a}{2y}$$

$$\frac{dy}{dx}(x_1, y_1) = \frac{3x_1^2 + a}{2y_1} = \lambda$$

If $P \neq -Q$, then

$$P + Q = (x_3, y_3)$$

$$x_3 = \lambda^2 - x_1 - x_2$$

$$y_3 = \lambda(x_1 - x_3) - y_1$$

where

$$\lambda = \begin{cases} \frac{y_2 - y_1}{x_2 - x_1} & \text{if } P \neq Q \\ \frac{3x_1^2 + a}{2y_1} & \text{if } P = Q \text{ (tangent)} \end{cases}$$

$(E, '+')$ is an abelian gp.

ELLIPTIC CURVE MODULO PRIME

$$E = \{ (x, y) \in \mathbb{R} \times \mathbb{R} \mid y^2 = x^3 + ax + b \}$$

$a, b \in \mathbb{R}$

define

elliptic curve points over $\mathbb{Z}_p$

elliptic curve over $\mathbb{Z}_p$,
where p is prime.

$$E = \{ (x, y) \in \mathbb{Z}_p \mid y^2 \equiv x^3 + ax + b \pmod{p} \}$$

$x \in \mathbb{Z}_p$ where $a, b \in \mathbb{Z}_p$

$$4a^3 + 27b^2 \not\equiv 0 \pmod{p} \} \cup \{ \emptyset \}$$

non singular elliptic curve over $\mathbb{Z}_p$.

$$P = (x_1, y_1) \in E$$

$$Q = (x_2, y_2) \in E$$

$$Q = -P, \quad P + (-P) = \emptyset$$

If $Q \neq -P$ then $P + Q = R(x_3, y_3) \in \mathbb{Z}_p \times \mathbb{Z}_p$

$$x_3 = \lambda^2 - x_1 - x_2 \pmod{p}$$

$$y_3 = \lambda(x_1 - x_3) - y_1 \pmod{p}$$

$$\lambda = \begin{cases} (y_2 - y_1)(x_2 - x_1)^{-1} \pmod{p} & \text{if } P \neq Q \\ (3x_1^2 + a)(2y_1)^{-1} \pmod{p} & \text{if } P = Q \end{cases}$$

$(E, +)$ is an abelian group with identity element $\emptyset$.

Q Perform additions

(i) $(2, 7) + (5, 2)$

(ii) $(3, 6) + (3, 6)$

(i) $(2, 7) \quad (5, 2)$
 $x_1 \ y_1 \quad x_2 \ y_2$

$$\begin{aligned} \lambda &= (y_2 - y_1)(x_2 - x_1)^{-1} \text{ mod } 17 \\ &= (2 - 7)(5 - 2)^{-1} \text{ mod } 17 \\ &= (-5) \text{ mod } 17 \cdot (3)^{-1} \text{ mod } 17 \\ &= (12 \times 6) \text{ mod } 17 \\ &= 72 \text{ mod } 17 \end{aligned}$$

$$\lambda = (4)$$

$$\begin{aligned} x_3 &= \lambda^2 - x_1 - x_2 \\ &= (16 - 2 - 5) \text{ mod } 17 \\ &= 9 \end{aligned}$$

$$\begin{aligned} y_3 &= (\lambda(x_1 - x_3) - y_1) \text{ mod } 17 \\ &= (4(2 - 9) - 7) \text{ mod } 17 \\ &= (-28 - 7) \text{ mod } 17 \\ &= (-35) \text{ mod } 17 \\ &= 16 \\ &\quad (9, 16) \end{aligned}$$

How we can find points on elliptic curve over $\mathbb{Z}_p$?

let $p = 11$

$$y^2 = x^3 + ax + b \pmod{11} \quad (x, y) \in \mathbb{Z}_p \times \mathbb{Z}_p$$

x	$x^3 + ax + b$	Quadratic residue mod p
0	y_1	}
1	y_2	
2	1	
3	1	
4	1	
5	1	
6	1	
7	1	
8	1	
9	1	
10	1	

Definition $x \in \mathbb{Z}_p$ is a quadratic ^{residue} modulo p

iff

$y^2 \equiv x \pmod{p}$ has a solution

$$y \in \mathbb{Z}_p$$

$$p = 11$$

$$(\pm 1)^2 = 1$$

$$y^2 \equiv x \pmod{p}$$

$$(\pm 5)^2 = 3$$

$$\underline{1, 3, 5, 9}$$

Euler's Criteria

Given an integer $x \in \mathbb{Z}_p$

$x \Rightarrow$ quadratic residue mod p iff

$$\boxed{(x^{(p-1)/2}) \equiv 1 \pmod{p}}$$

$$y^2 = x$$

$$y \in \mathbb{Z}_p \quad x^{(p-1)/2} = y^{p-1} \equiv 1 \pmod{p}$$

Theorem Suppose z is quadratic residue mod p

$$\text{and } \boxed{p \equiv 3 \pmod{4}}$$

then two square roots of $z \pmod{p}$ are

$$\pm z^{(p+1)/4} \pmod{p} \rightarrow z^{(1+1)/4} \pmod{11}$$

$$\text{Ex} = p = 11$$

$$y^2 = x^3 + x + 6 \text{ over } \mathbb{Z}_{11}$$

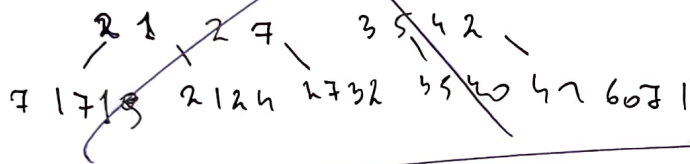
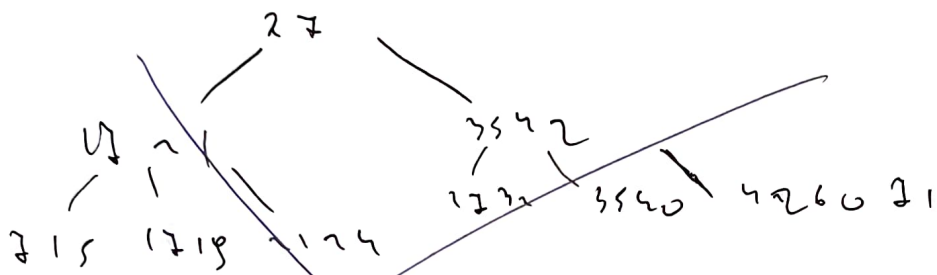
$$x \in \mathbb{Z}_{11}$$

$$a = 1$$

$$b = 6$$

$$y^{(p-1)/2}$$

x	y $x^3 + x + 6 \pmod{11}$	Quadratic residue	Square roots
0 ✓	6	x	
1 ✓	8	x	
2 ✓	$8 + 2 + 6 \pmod{11} = 5$	✓	$5^3 \pmod{11} = 4, 7$ ✓
3 ✓	3	✓	5, 6 ✓
4 ✓	8	x	
5 ✓	4	✓	9, 2 ✓
6 ✓	9	x	
7 ✓	4	✓	9, 2 ✓
8 ✓	9	✓	3, 8 ✓
9 ✓	7	x	
10	4	✓	9, 2 ✓ 12^9 6



points are

- $(2, 4)$ $(2, 7)$
- $(3, 5)$ $(3, 6)$
- $(5, 2)$ $(5, 9)$
- $(7, 9)$ $(7, 2)$
- $(8, 3)$ $(8, 8)$
- $(10, 9)$ $(10, 2)$



Q $y^2 \equiv x^3 + 2x + 2 \pmod{17}$

x	$(x^3 + 2x + 2) \pmod{17}$	Quadratic residue	
0	2 ✓	✓	✓ (0, 6) (0, 11)
1	5 ✗		✓ (7, 6) (7, 11)
2	14 ✗		
3	1 ✓	✓	✓ (3, 11) (3, 16)
4	6 ✗		✓ (9, 11) (9, 16)
5	1 ✓	✓	✓ (5, 11) (5, 16)
6	9 ✓	✓	✓ (6, 3) (6, 14)
7	2 ✓	✓	✓ (10, 6) (10, 11)
8	3 ✗		✓ (13, 7) (13, 10)
9	1 ✓	✓	✓ (16, 13) (16, 14)
10	2 ✓	✓	
11	12 ✗		
12	3 ✗		
13	15 ✓	✓	
14	3 ✗		
15	7 ✗		
16	16 ✓	✓	

$x^2 \pmod{17}$

x	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$x^2 \pmod{17}$	1	4	9	16	8	2	15	13	13	15	2	8	16	9	4	1

Q

$$y^2 = x^3 + 3x + 2$$

over $\mathbb{Z}_7$

$$a = 3$$

$$b = 2$$

compute all points over $\mathbb{Z}_7$

x	$x^3 + 3x + 2 \pmod{7}$	residue modulo	$(p+1)/4$
0	2	✓	1, 6 $(2^{(7+1)/4} \pmod{7}, 3)$
1	6	✗	
2	2	✓	1, 6 4, 3
3	3	✗	
4	1	✓	1, 6 , ✓
5	2	✓	1, 6 (4, 3)
6	5	✗	

$$(0, 4) \quad (0, 3) \quad (2, 4) \quad (2, 3)$$

$$0 \quad (4, 1) \quad (4, 6) \quad (5, 4) \quad (5, 3)$$

$$\underline{0(4) = 9}$$

Q005